$$\langle \Psi | \Phi \rangle = \int_{-\infty}^{\infty} \psi^*(x)\phi(x)dx. \qquad (8.2)$$

Probability Densities Replace Probabilities: Later, we will identify

$$P(x) = \psi^*(x)\psi(x)$$

as a probability density for the variable *x*. Why a probability density and not just a probability? If *x* is a continuous variable, then the probability that it will have any exact value is typically zero. A more useful question to ask is: What is the probability that *x* lies between two values, $x = a$ and $x = b$? Probability densities are defined so that this probability is given by an integral:

$$P(a,b) = \int_a^b P(x)\ dx = \int_a^b \psi^*(x)\psi(x)\ dx.$$

Because the total probability should be 1, we can define a normalized vector by

$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)\ dx = 1. \qquad (8.3)$$

Dirac Delta Functions Replace Kronecker Deltas: So far, this should be very familiar. The Dirac delta function may be less so. The delta function is the analog of the Kronecker delta, δ_{ij}. The Kronecker delta is defined to be 0 for $i \neq j$ and 1 for $i = j$. But it can also be defined another way. Consider any vector F_i in a finite dimensional space. It is easy to see that the Kronecker delta satisfies